EGR304 — Power Budget Report

Subsystem: User Controls & Safety

Team: 107

Microcontroller: PIC18F57Q43 Curiosity Nano

Report Status: Preliminary Design — Pending Hardware Verification

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Overview

1. Introduction

The purpose of this Power Budget is to estimate the electrical power requirements of the User Controls & Safety subsystem before hardware assembly and testing.

The subsystem includes a PIC18F57Q43 microcontroller, emergency stop button, start/pause button, speed control potentiometer, green and red indicator LEDs, and an audible warning buzzer.

The power budget evaluates component current consumption, power distribution, voltage regulator capacity, and thermal performance. A 25% safety margin is applied to the estimated load currents to provide additional design headroom.

2. Power Distribution Architecture

The subsystem uses two separate power sources.

The PIC18F57Q43 Curiosity Nano operates from USB power with a target voltage of 5V and a system clock frequency of 64MHz. The microcontroller, indicator LEDs, speed potentiometer, and buzzer transistor base drive are assigned to the USB-powered target rail.

A separate 9V DC adapter supplies an L7805CV linear voltage regulator, which provides 5V to the piezoelectric buzzer and a proposed minimum-load resistor.

Both power systems share a common signal ground, while their 5V supply outputs remain electrically isolated from each other.

3. Preliminary Results

Parameter	Estimated Value
External 5V Rail Current	10.49mA
External 5V Rail with 25% Margin	13.12mA
Estimated 9V Input Current	21.12mA
USB Target Rail Current	40.81mA
USB Target Rail with 25% Margin	51.01mA
L7805CV Power Dissipation	0.124W
Estimated Junction Temperature at 25°C	31.22°C

These values are preliminary design estimates based on selected component specifications and assumed operating conditions. Final verification will be performed after hardware assembly.

Component Power Requirements

	A	B	C	D	E	F	G	H
8	**All Major Components**	**Component Name**	**Part Number**	**Supply Voltage / Conditions**	**Qty.**	**Preliminary / verified current (mA)**	**Total Current (mA)**	**Unit**
9		Microcontroller	PIC18F57Q43	USB 5V, 64MHz	1	20	20	mA
10		Passive Piezo Buzzer	CPT-1407-5TH	External +5V	1	5	5	mA
11		Buzzer Driver	2N3904TF	RB4 GPIO drive	1	0.91	0.91	mA
12								
13								
14								
15								
16	***External +5V Power Rail***	**Component Name**	**Part Number**	**Supply Voltage Range**	**Qty.**	**Budget current (mA), provisional**	**Total Current (mA)**	**Unit**
17		Passive Piezo Buzzer	CPT-1407-5TH	5V drive	1	5	5	mA
18		910Ω preload (planned)	910Ω, 1%	External +5V	1	5.49	5.49	mA
19								
20						***Subtotal***	10.49	mA
21						***Safety Margin***	25%	
22						***Required current (+25%)***	13.12	mA
23								
24	***Regulator***	5V Linear Regulator	L7805CV	9V to 5V	1	1500	1500	mA
25						***Regulator rated headroom***	1486.88	mA

4. Major Component Analysis

The power consumption of each active component was evaluated to estimate

the subsystem's total current requirements.

The PIC18F57Q43 operates at 5V with a system clock frequency of 64MHz. A preliminary current allowance of 20mA was assigned for design calculations. This value is a planning assumption rather than a verified maximum current from the manufacturer.

The selected CPT-1407-5TH piezoelectric buzzer has a nominal current consumption of approximately 5mA under specified operating conditions.

A 2N3904 NPN transistor is used as the proposed buzzer switching device. Its base current is determined by the microcontroller output voltage and the selected base resistor.

The 10kΩ speed control potentiometer and the two indicator LEDs are also included in the power calculations.

5. External 5V Power Rail

The external 5V rail is supplied by the L7805CV regulator and powers the piezoelectric buzzer and a proposed 910Ω, 1% minimum-load resistor.

The minimum-load resistor is included to help satisfy the L7805CV's specified minimum-load operating condition when the buzzer is inactive.

The estimated current consumption is:

- Buzzer: 5.00mA
- Minimum-load resistor: 5.49mA
- Total: 10.49mA

Including the required 25% safety margin:

Required Current = 10.49mA × 1.25 = 13.12mA

The selected regulator has adequate nominal current capacity for this preliminary load. Its actual operating capability must also satisfy thermal limitations.

Figure 1. Major component current requirements and external 5V power rail

calculations.

USB Power Rail

	A	B	C	D	E	F	G	H
27	***USB +5V Rail (Target)***	**Component Name**	**Part Number**	**Supply Voltage Range**	**Qty.**	**Budget current (mA), provisional**	**Total Current (mA)**	**Unit**
28		PIC target (64 MHz)	PIC18F57Q43	VTG = 5V	1	20.00	20.00	mA
29		Red + green LEDs	330 Ω / Vf,min 1.8V	GPIO = 5V	2	9.70	19.39	mA
30		Pot + BJT base	10k pot + 4.7k base	VTG = 5V	1	1.41	1.41	mA
31						***Subtotal***	40.81	mA
32						***Safety Margin***	25%	
33						***Required Current (incl. 25%)***	51.01	mA
34								
35	***Regulator***	Nano target LDO	MIC5353	USB → VTG	1	500	500.00	mA
36						***Remaining Regulator Rated Current***	448.99	mA
37								
38	***3.3V Rail — N/A***	**Component Name**	**Part Number**	**Supply Voltage Range**	**Qty.**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
39								mA
40							0	mA
41							***Subtotal***	mA
42							***Safety Margin***	
43							***Total Current Required on +3.3V Rail***	mA
44								
45	***Regulator***							mA
46								mA

6. USB-Powered 5V Rail

The USB-powered 5V rail supplies the PIC18F57Q43 microcontroller and the associated user control components.

The PIC operating current is provisionally estimated at 20mA for planning purposes.

Each indicator LED uses a proposed 330Ω current-limiting resistor. Assuming a minimum LED forward voltage of 1.8V, each LED draws approximately 9.70mA. Both LEDs are assumed to operate simultaneously for a conservative load estimate.

The 10kΩ speed potentiometer draws approximately 0.50mA at 5V.

The 2N3904 transistor base current is estimated at 0.91mA using a proposed 4.7kΩ base resistor and an assumed base-emitter voltage of 0.7V.

The total estimated USB target load is **40.81mA**.

After applying the 25% safety margin:

Required USB Target Current = 40.81mA × 1.25 = 51.01mA

This value represents the estimated target circuit current and does not include the Curiosity Nano debugger or other onboard USB power consumption.

Figure 2. USB-powered 5V target rail and estimated current requirements.

External Power Sources

	A	B	C	D	E	F	G	H
48	***9V DC source***	**Component Name**	**Part Number**	**Supply Voltage Range**	**Output Voltage**	**Rated supply capacity (mA)**	**Total Current (mA)**	**Unit**
49	**Power Source 1 Selection**	9V DC Wall Adapter	Tensility 16-00134	100–240V AC	9V	2000	2000	mA
50								
51	**Power Rails Connected to External Power Source 1**	5V Linear Regulator	L7805CV	9V Input	1	21.12	21.12	mA
52								mA
53								mA
54						***Remaining rated supply current***	1978.88	mA
55								
56	***USB +5V Power Source***	**Component Name**	**Part Number**	**Supply Voltage Range**	**Output Voltage**	**Rated / provisional current (mA)**	**Total Current (mA)**	**Unit**
57	**Power Source 2 Selection**	USB 5V (PC/debug USB)	USB connection	nominal 5V	5V			mA
58								
59	**Power Rails Connected to External Power Source 2**	Nano target + USB board ov	Curiosity Nano debug	USB input curre	1			mA
60								
61								
62						***Remaining USB current (pending)***		mA

7. External 9V Power Supply

The selected external power source is the Tensility International 16-00134, providing a regulated 9V DC output with a maximum rated current of 2A.

The 9V supply powers the L7805CV voltage regulator, which converts the input voltage into a regulated 5V output.

The estimated regulator input current includes the required external rail current and a provisional regulator quiescent current of 8mA.

Estimated Input Current = 13.12mA + 8mA = 21.12mA

The selected 9V adapter provides sufficient rated current capacity for the estimated load.

8. USB Power Source

The PIC18F57Q43 Curiosity Nano is powered through its USB connection.

The current budget for the microcontroller and target components has been

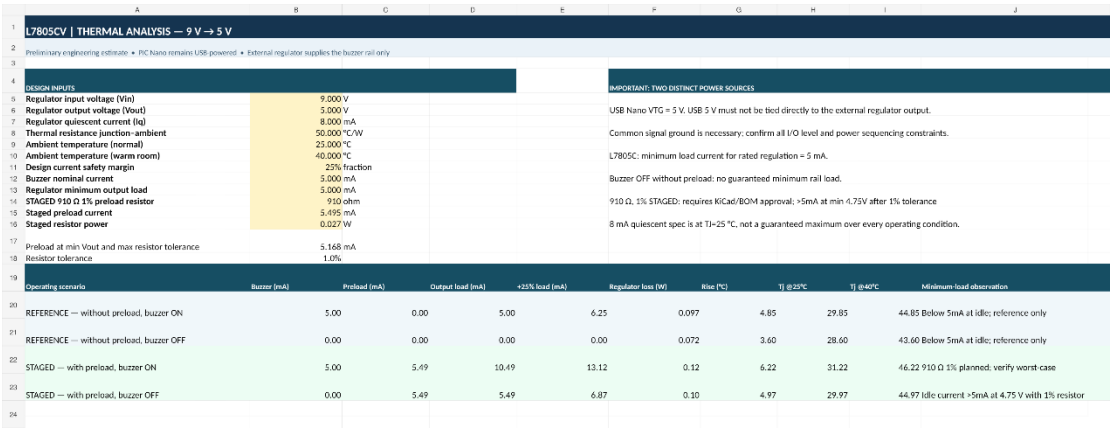
estimated separately from the external 9V power system.

However, the total USB input current cannot yet be determined because the onboard debugger current and available USB source current have not been fully verified.

These values will be confirmed during hardware testing.

Figure 3. External 9V power source and USB supply capacity analysis.

Thermal Analysis and Conclusions



L7805CV | THERMAL ANALYSIS — 9 V → 5 V

Preliminary engineering estimate • PIC Nano remains USB-powered • External regulator supplies the buzzer rail only

DESIGN INPUTS		IMPORTANT: TWO DISTINCT POWER SOURCES
Regulator input voltage (Vin)	9.000 V	
Regulator output voltage (Vout)	5.000 V	USB Nano VTG = 5 V. USB 5 V must not be tied directly to the external regulator output.
Regulator quiescent current (Iq)	8.000 mA	
Thermal resistance junction-ambient	50.000 °C/W	Common signal ground is necessary; confirm all I/O level and power sequencing constraints.
Ambient temperature (normal)	25.000 °C	
Ambient temperature (warm room)	40.000 °C	L7805C: minimum load current for rated regulation = 5 mA.
Design current safety margin	25% fraction	
Buzzer nominal current	5.000 mA	Buzzer OFF without preload: no guaranteed minimum rail load.
Regulator minimum output load	5.000 mA	
STAGED 910 Ω 1% preload resistor	910 ohm	910 Ω, 1% STAGED: requires KiCad/BOM approval; >5mA at min 4.75V after 1% tolerance
Staged preload current	5.495 mA	
Staged resistor power	0.027 W	8 mA quiescent spec is at TJ=25 °C, not a guaranteed maximum over every operating condition.
Preload at min Vout and max resistor tolerance	5.168 mA	
Resistor tolerance	1.0%	

Operating scenario	Buzzer (mA)	Preload (mA)	Output load (mA)	+25% load (mA)	Regulator loss (W)	Rise (°C)	TJ @25°C	TJ @40°C	Minimum-load observation
REFERENCE — without preload, buzzer ON	5.00	0.00	5.00	6.25	0.097	4.85	29.85	44.85	Below 5mA at idle; reference only
REFERENCE — without preload, buzzer OFF	0.00	0.00	0.00	0.00	0.072	3.60	28.60	43.60	Below 5mA at idle; reference only
STAGED — with preload, buzzer ON	5.00	5.49	10.49	13.12	0.12	6.22	31.22	46.22	910 Ω 1% planned; verify worst-case
STAGED — with preload, buzzer OFF	0.00	5.49	5.49	6.87	0.10	4.97	29.97	44.97	Idle current >5mA at 4.75 V with 1% resistor

9. L7805CV Thermal Analysis

The L7805CV linear voltage regulator converts the external 9V input into a regulated 5V output.

Because a linear regulator dissipates excess electrical energy as heat, its power dissipation and junction temperature must be evaluated.

The estimated power dissipation is calculated using:

P = (Vin – Vout) × Iout + Vin × Iq

Using a 9V input, 5V output, 13.12mA budgeted load current, and 8mA provisional quiescent current:

Estimated Power Dissipation = 0.124W

Assuming a junction-to-ambient thermal resistance of 50°C/W:

Temperature Rise = 0.124 × 50 = 6.22°C

At an ambient temperature of 25°C:

Estimated Junction Temperature = 31.22°C

At an ambient temperature of 40°C:

Estimated Junction Temperature = 46.22°C

These estimates indicate that the L7805CV is expected to operate with a relatively low thermal load under the proposed conditions.

Additional heatsinking is not expected to be necessary for this preliminary design. Final thermal performance will be verified after PCB assembly.

Figure 4. L7805CV power dissipation and junction temperature estimation.

10. Conclusions

The preliminary Power Budget indicates that the selected power distribution system has sufficient nominal current capacity to support the User Controls & Safety subsystem under the stated design assumptions.

The 9V DC adapter and L7805CV linear voltage regulator are expected to provide adequate power for the external buzzer circuit. The estimated regulator power dissipation is low, suggesting that additional thermal management may not be required.

The USB-powered target rail is estimated to require approximately 51.01mA, including the 25% current safety margin. However, the complete USB input current remains unverified.

Several component values, including the PIC operating current, LED forward voltage, buzzer current, and transistor base resistance, are based on preliminary design assumptions.

After hardware assembly, actual voltage, current consumption, buzzer performance, and regulator temperature will be measured to validate and refine the Power Budget.

Overall, the proposed power architecture appears feasible for the subsystem, subject to successful hardware verification.

11. References

1. Microchip Technology — *PIC18F27/47/57Q43 Datasheet*
2. Microchip Technology — *PIC18F57Q43 Curiosity Nano Hardware User Guide*
3. STMicroelectronics — *L78 Series Positive Voltage Regulators Datasheet*
4. Same Sky — *CPT-1407-5TH Piezoelectric Buzzer Product Specifications*
5. EGR304 — *Component Selection and Power Budget Assignment Requirements*

Report Status: Preliminary — Hardware Verification Pending